K. Rajitha
192524149

Assignment II

SDG-Based Questions – CO2

Course Code	ITA06
Course Name	Machine Learning
Faculty Handling	
Course Outcome	CO2 – Neural Networks and Genetic Algorithms

Instructions: Answer any question(s) as directed by the faculty. Each question maps a Neural Network / Genetic Algorithm concept to a real-world application aligned with the United Nations Sustainable Development Goals (SDGs).

Q.No	Question
1	Design a single-layer perceptron that classifies water samples as 'safe' or 'unsafe' for drinking based on pH, turbidity, and dissolved oxygen levels. Explain the weight update rule you would use. (SDG 6 – Clean Water and Sanitation)
2	A perceptron is trained to classify farmland soil as 'fertile' or 'degraded' using nitrogen, phosphorus, and potassium content as inputs. Derive the perceptron learning rule and explain convergence conditions. (SDG 2 – Zero Hunger)
3	Explain why a single perceptron cannot solve the problem of classifying households as 'energy poor' or 'energy secure' when the decision boundary is non-linear. Illustrate with a suitable example. (SDG 7 – Affordable and Clean Energy)
4	Using the perceptron algorithm, design a binary classifier that predicts whether a patient is at risk of malnutrition based on BMI and dietary intake. Show two iterations of weight updates with sample data. (SDG 2 – Zero Hunger)
5	Discuss the representational limitations of perceptrons in the context of classifying air quality as 'hazardous' or 'safe' when pollutant interactions are non-linear. (SDG 11 – Sustainable Cities and Communities)

Single-layer Perceptron for water Quality

A single-layer perceptron classifies water as safe (1) or unsafe (0) using pH, turbidity, and dissolved oxygen.

Inputs:

- x_1 : PH
- x_2 : Turbidity
- x_3 : Dissolved oxygen

Perceptron model:

$$y = f(w_1x_1 + w_2x_2 + w_3x_3 + b)$$

Here, w represents weights, b is bias, and f is the step activation function.

weight updata rule:

$$w_i(new) = w_i(old) + \eta(y - \hat{y})x_i$$

$$b(new) = b(old) + \eta(y - \hat{y})$$

where η is the learning rate, y is the actual Output, and $\hat{y}$ is the predicted Output.

Conclusion: The perceptron learns from labelled water samples and helps classify quality. Actual drinking-water safety

2) Perceptron for Soil Fertility

A Perceptron classifies farmland soil as Fertile (1) or Degraded (0) using nitrogen, phosphorus, and Potassium (NPK) levels.

Learning rule

$$\hat{y} = f(w_1x_1 + w_2x_2 + w_3x_3 + b)$$

$$w_i(new) = w_i(old) + \eta(y - \hat{y})x_i$$

$$b(new) = b(old) + \eta(y - \hat{y})$$

Convergence Conditions

1) Training samples must be correctly labelled
2) Data should be linearly separable.
3) A suitable positive learning rate must be selected.
4) weights are updated whenever a prediction is incorrect.
5) The Perceptron Convergence theorem guarentee Convergence in a finite number of updates for linearly separable data.

Conclusion:

The perceptron can classify soil fertility and support better agricultural decisions.

Limitations of a Single Perceptron

A single perceptron can classify only linearly separable data. It cannot learn complex, non-linear decision boundaries.

Examples: classifying households as Energy-Poor or Energy-secure based on electricity access and income may require a non-linear boundary when the combined effects of these features are complex.

The perceptron uses:

$$y = f(w_1x_1 + w_2x_2 + b)$$

this Equation cretes a straight decision boundary.

Limitations

1) It cannot Represent Complex non-linear relationships.
2) It may fail when classes Overlap.
3) It cannot solve the XOR classification Problem.
4) Its prediction depends on a linear combination of inputs.

Solution: use a multilayer perceptron with hidden layers non-linear activation functions such as ReLU.

Conclusion: multi-layer neural networks are some suitable for complex household Energy classification

24) Perceptron for Malnutrition Risk

A perceptron predicts whether a Patient is at Risk (1) or Not a Risk (0) using BMI and dietary intake.

Assume:-

- Initial weights : $w_1 = 0.2$, $w_2 = 0.3$
- Bias : $b = 0$
- Learning rate : $\eta = 0.1$
- Inputs are normalized between 0 & 1.

Update formula:

$$w_i(new) = w_i(old) + \eta (y - \hat{y}) x_i$$

Two Iterations

Iteration	x_1, x_2	Actual y	Predicted $\hat{y}$
1	(0.2, 0.3)	1	0
2	(0.8, 0.9)	0	1

Iteration 1 :-

$$\text{Error} = 1 - 0 = 1$$

$$w_1 = 0.2 + 0.1(1)(0.2) = 0.22$$

$$w_2 = 0.3 + 0.1(1)(0.3) = 0.33$$

$$b = 0 + 0.1(1) = 0.1$$

Iteration 2:

$$\text{Error} = 0 - 1 = -1$$

$$w_1 = 0.22 + 0.1(-1)(0.8) = 0.14$$

$$w_2 = 0.33 + 0.1(-1) = 0$$

Final weights :- $w_1 = 0.14$,

$w_2 = 0.24$,

$b = 0$

Conclusion: The perceptron updates its weights using prediction errors to improve malnutrition-risk classification.

5. Limitations of Perceptrons in Air Quality classification :-

A single perceptron classifies air quality as safe or Hazardous using Pollutants such as $PM_{2.5}$, PM_{10}, and nitrogen dioxide (NO_2).

Its output is:

$$y = f(w_1 x_1 + w_2 x_2 + w_3 x_3 + b)$$

Representational limitations:-

1) Linear boundary: It cannot represent complex non-linear decision boundaries.
2) Pollutant interactions: Different Pollutants may interact and Jointly affect air quality.
3) complex Patterns: It cannot easily learn sudden changes or complicated Popllution patterns.
4) overlapping classes: Similar Pollutant values may occur in both safe and hazardous samples.

Solution: Use a multilayer perceptron, Random forest, or another suitable non-linear model. Train and Evaluate it using real air-quality data.